

Qin Yang

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CERTIFICATES: *Power BI Data Analyst Associate* | *Snowflake SnowPro Core* | *Microsoft Azure Data Scientist Associate*

Summary

Data Scientist with a proven record of turning large-scale, complex data into statistically rigorous, actionable insight across marketing, engineering, and financial-services environments. Skilled in experimental design, hypothesis testing, and causal inference (A/B testing); end-to-end ML and LLM/NLP solution development (RAG, chatbots, time-series forecasting); and building scalable, production-grade data pipelines (Snowflake, Databricks, Azure ML) with MLOps discipline. Experienced translating model outputs into enterprise BI and business-intelligence reporting for VP-level stakeholders, and comfortable driving ambiguous, cross-functional initiatives to completion.

Experience

Data Scientist, Marketing @Tractor Supply Co.

Brentwood, TN | 2025 Jan - Present

- **Statistical Analysis & Experimental Design:** Designed and executed A/B tests with rigorous statistical analysis and hypothesis testing to quantify marketing-campaign effectiveness; translated results into clear, actionable recommendations for cross-functional stakeholders across Marketing, Finance, and Merchandising.
- **LLM Deployment:** Built a pipeline leveraging LLMs (e.g., Claude Sonnet, GPT-OSS) in Databricks to process product information and automate classification tasks such as product category identification — including prompt design and output quality evaluation.
- **Large-Scale Data Pipelines:** Engineered scalable SQL pipelines integrating 1M+ daily records (site traffic, financial) into Snowflake, creating analytically-ready, quality-validated datasets for downstream modeling and reporting.
- **ML Model Development & Deployment:** Built purchase-propensity models using XGBoost on Databricks, with MLflow for experiment tracking and model versioning; outputs feed downstream recommendation systems that identify high-intent customer segments.
- **Business Intelligence:** Built automated Power BI dashboards using Snowflake SQL and Git-based workflows, enabling real time KPI monitoring and replacing manual reporting processes.
- **Stakeholder Communication:** Partnered with VP-level leadership to define success metrics and standardize reporting frameworks, ensuring analytical outputs directly informed budget-allocation decisions.

Data Scientist, Global Automation & Engineering @Assurant

Nashville, TN | 2023 Feb - Jan 2025

- **Predictive Modeling & MLOps:** Designed and productionized an XGBoost classification model to automatically identify mobile device models, integrating it into a Dockerized CI/CD pipeline (Azure ML, MLflow) that cut manual identification time and accelerated the downstream device-processing workflow.
- **Process Automation (NLP + Computer Vision):** Architected an NLP + computer-vision system that reads device screens and replicates human interaction patterns to autonomously operate and test mobile devices — eliminating manual handling, cutting labor costs by \$300K annually, and unlocking a new external revenue stream from the automated service.
- **KPI Dashboarding & Performance Monitoring:** Designed and built real-time dashboards surfacing core production KPIs (devices processed, error rate, pass rate), giving factory leadership and floor operators continuous visibility into throughput and model reliability and enabling faster response to processing anomalies.
- **Stakeholder Communication & Advisory:** Reported directly to VP-level leadership, translating model performance and KPI trends into operational guidance that directly shaped day-to-day factory workflow and process decisions.
- **ETL & Data Engineering:** Built and maintained ETL pipelines pulling from internal and external APIs and databases, cleaning and transforming data into centralized, quality-monitored repositories that fed both the classification model and the KPI dashboards.

Research Assistant (Automatic Speech Recognition, ASR), Center for Data Science@NYU

New York, NY | 2022 Sep - 2023 May

- **Model Development:** Fine-tuned an ASR model on 375 hours of labeled and pseudo-labeled audio data using PyTorch; integrated a word-based language model, reducing word error rate to 13.5%.
- **Reproducible ML Workflows:** Tracked experiments using Weights & Biases and high-performance computing (HPC) to ensure reproducible workflows and accelerated training.
- **Research & Technical Writing:** Co-authored and published research at the 2023 *Interspeech Conference*, demonstrating strong technical writing and communication of data-driven findings.

Data Science Intern, Digital Transformation @Campana & Schott

New York, NY | 2022 May - 2022 Sep

- **Data Pipelines:** Built robust ETL pipelines to extract data from APIs and databases, transform/clean it, and load it into centralized repositories for modeling and analysis.
- **Applied NLP & Forecasting:** Applied NLP (NER, sentiment analysis) and time-series models (LSTM, N-HiTS) to extract insight from unstructured pharmaceutical data, supporting strategic planning.
- **Business Intelligence:** Developed Power BI dashboards to visualize KPIs and trends, enabling stakeholders to monitor business metrics and make data-informed decisions

Education

- Georgia Institute of Technology

2025 Sep - Present

- Master's Degree in Computer Science (*Online*)
 - Relevant Courses: Machine Learning for Trading, Database System Concept and Design, Data and Visual Analytics
- New York University

2021 Sep - 2023 May

- Master's Degree in Data Science
 - Relevant Courses: Machine Learning, Applied ML in Finance, NLP and Computational Semantics, Deep Learning, Time Series Analysis, Probability & Statistics, Big Data
- Shandong University

2014 Sep - 2021 May

- Master's Degree in Economics/Bachelor's Degree in Management Science & Engineering
 - Relevant Courses: Advanced Micro & Macro Econ, Econometrics, Finance, Calculus, Linear Algebra, Data Principles and Applications, Python and Data Analysis, Database Management

Skills & Tools

- **Statistics & Experimentation:** A/B Testing, Experimental Design, Hypothesis Testing, Causal Inference
- **AI / LLM / NLP:** LLM Fine-Tuning & Evaluation, RAG, Chatbots, LangChain, Text Classification, Sentiment Analysis, NER, ASR
- **Forecasting & ML:** Time Series (LSTM, N-HiTS), Regression, Classification & Clustering, Recommendation Systems (RNN/DNN/Collaborative Filtering)
- **Programming Languages:** Python, SQL
- **MLOps & Cloud:** Azure ML, MLflow, Docker, CI/CD, Databricks, Version Control (Git), Reproducible Pipelines
- **Data Engineering & Warehousing:** Snowflake, ETL, PySpark, PostgreSQL, MySQL, MongoDB, Vector DB, Azure Cosmos DB
- **Visualization & BI:** Power BI, Plotly, Dash, Streamlit
- **Frameworks:** PyTorch, TensorFlow, scikit-learn, FastAPI, Flask, Django
- **Other:** RESTful API, Multithreading/Multiprocessing, Linux, HTML/CSS, JIRA, Adobe Analytics, Supply Chain & Marketing Analytics
- **Soft Skills:** Stakeholder Engagement, Mentoring & Knowledge Sharing, Cross-Team Collaboration, Project Management, Ambiguity-to Execution, Technical Writing